Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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Name: D.Kanishkarane
Reg No: 24011101021

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

Dataset used

All 3,670 images of `tf_flowers` (original `flower_photos.tgz`) were used, with labels in alphabetical order (daisy, dandelion, roses, sunflowers, tulips). Figure 1 shows samples of every class (required visualization 1).

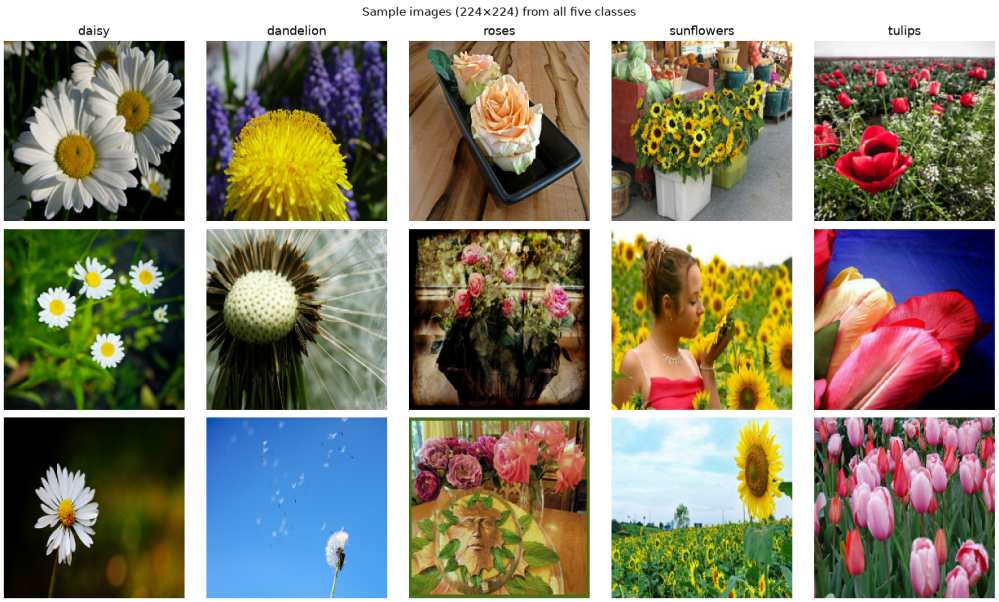


Figure 1: Sample input images (224×224) from all five classes.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.

Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

Preprocessing and augmentation used

Images were resized once to $224 \times 224 \times 3$ and passed through `resnet50.preprocess_input` (BGR, ImageNet mean subtraction). Training batches only were augmented with the same seeded recipe for every model: `RandomFlip("horizontal")`, `RandomRotation(0.05)` ($\approx \pm 18°$), `RandomZoom(0.1)`, `RandomTranslation(0.1, 0.1)` and `RandomContrast(0.1)`.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

Split used

One stratified split (seed 42) was saved to disk and reused by all 17 trained models (Table 1).

Table 1: Images per class in the fixed stratified 70 : 15 : 15 split.

Subset	daisy	dandelion	roses	sunflowers	tulips	**Total**	**Share**
Train	443	629	449	489	559	2,569	70 %
Val	95	134	96	105	120	550	15 %
Test	95	135	96	105	120	551	15 %
All	633	898	641	699	799	3,670	100 %

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

Verification

These dimensions were confirmed on the Keras ImageNet ResNet50 (Table 2). conv5_x holds 63.6 % of the 23.59 M backbone parameters.

Table 2: ResNet50 stages measured on the Keras ImageNet model (input $224 \times 224 \times 3$).

Stage	Output layer	$H \times W$	C	Parameters
Stem + MaxPool	`pool1_pool`	56×56	64	9,728
conv2_x	`conv2_block3_out`	56×56	256	220,032
conv3_x	`conv3_block4_out`	28×28	512	1,230,336
conv4_x	`conv4_block6_out`	14×14	1024	7,129,088
conv5_x	`conv5_block3_out`	7×7	2048	14,998,528
Total backbone				23,587,712

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

Observation

In Figure 2, the early maps (56×56) still show tulip-head outlines and textures. conv3_x responds to individual tulip heads (flower parts), and conv4_x/conv5_x show only coarse class-evidence regions.

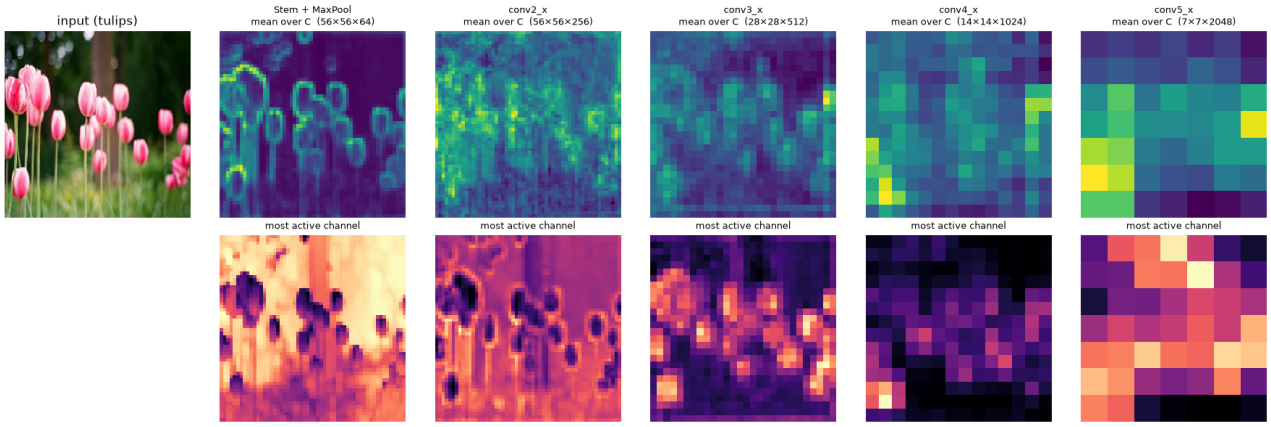


Figure 2: Output of every ResNet50 stage for one test image: mean over channels (top) and most active channel (bottom).

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.

Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.

Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

Which attention mechanism should be used?

and

Where should the attention mechanism be inserted?

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$P_2, \quad P_3, \quad P_4$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

Implementation note

ResNet50 was split at `conv2_block3_out`, `conv3_block4_out`, `conv4_block6_out` and `conv5_block3_out` into segments that share the pretrained layers. P_1 was trained once (B1) to complete the position study.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

Baseline result

The baseline (A0, Dropout 0.3, 23,597,957 parameters) reached **accuracy 0.9437** (520/551) and **macro F1 0.9436** on the test set, with 12.85 ms per image.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2\,\delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

Which feature channels are important?

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

Implementation

GAP $\rightarrow$ `Dense`$(C/16)$+ReLU $\rightarrow$ `Dense`(C)+sigmoid, multiplied with F. At P_3 this adds 132,160 parameters. This is the SE computation, so it is reported as "Channel / SE".

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \,\|\, F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F_s' = F \odot M_s(F).$$

Spatial attention answers:

Which spatial locations are important?

Implementation

One `Conv2D`$(1, k\times k)$+sigmoid on $[F_{\text{avg}}\|F_{\max}]$. With $k = 7$ it adds 99 parameters at any position.

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

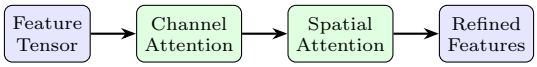
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



Implementation

The two modules are chained in a configurable order: channel→spatial here, spatial→channel in Experiment 4. At P_3 this adds 132,259 parameters.

12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$
$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\, \text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

Implementation and cost

The module uses $d = C/8$ for Q and K, and V keeps C channels. γ is initialised at 0 and learned 0.0664. At P_3 it adds 1,310,721 parameters. The size of A grows 16× each time the resolution doubles (Table 3), which confirms that P_3 or P_4 should be used.

Table 3: Size of the self-attention matrix A at each candidate position.

Position	$H \times W$	$N = HW$	Entries of A (N^2)	A per image (MB, fp32)	A per batch of 32 (MB)
P_1	56×56	3,136	9,834,496	39.338	1,259
P_2	28×28	784	614,656	2.459	79
P_3	14×14	196	38,416	0.154	4.92
P_4	7×7	49	2,401	0.010	0.31

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

Recorded feature dimensions

The builder logged and asserted the shape before and after every attention block. No module changed (H, W, C) (Table 4). Table 5 lists the parameters each module adds.

Table 4: Feature dimensions immediately before and after the attention block (identical for every module).

Position	Shape before	Shape after	Experiments using this position
P_1	(56, 56, 256)	(56, 56, 256)	B1
P_2	(28, 28, 512)	(28, 28, 512)	B2, C4
P_3	(14, 14, 1024)	(14, 14, 1024)	A1, A2, A3, A4, A5, C3, C4, D1, E1, E2, F1, F2, F3
P_4	(7, 7, 2048)	(7, 7, 2048)	B4, C3, C4

Table 5: Parameters added by each attention module at each position (every module preserves (H, W, C)).

Module	P_1 (56×56×256)	P_2 (28×28×512)	P_3 (14×14×1024)	P_4 (7×7×2048)
Channel (= SE)	8,464	33,312	132,160	526,464
Spatial (7×7)	99	99	99	99
Channel→Spatial	8,563	33,411	132,259	526,563
Spatial→Channel	8,563	33,411	132,259	526,563
Self-attention	81,921	327,681	1,310,721	5,242,881
Combined (C→S→SA)	90,484	361,092	1,442,980	5,769,444
ECA	5	5	5	7
CBAM	8,563	33,411	132,259	526,563
Coordinate	6,696	25,680	100,512	397,632

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

Results

Table 6: Experiment 1 – attention type at P_3 (test set, 551 images). Best values in bold.

Model	Accuracy	Precision	Recall	Macro F1	ΔAcc (pp)	Attention params	Total params	Inference (ms)
A0: Baseline ResNet50	**0.9437**	0.9489	0.9409	**0.9436**	+0.00	0	23,597,957	12.85
A1: Channel Attention	0.9183	0.9218	0.9178	0.9192	−2.54	132,160	23,730,117	18.86
A2: Spatial Attention	0.9347	0.9379	0.9312	0.9340	−0.91	99	23,598,056	15.78
A3: Channel+Spatial	0.9238	0.9286	0.9212	0.9238	−2.00	132,259	23,730,216	18.56
A4: Self-Attention	0.9419	0.9438	0.9392	0.9411	−0.18	1,310,721	24,908,678	17.03
A5: Combined Attention	0.9111	0.9120	0.9104	0.9102	−3.27	1,442,980	25,040,937	17.24

Observations and inference

- No attention type at P_3 beat the baseline (0.9437). Self-attention (-0.18 pp) and spatial attention (-0.91 pp) were closest; Combined was lowest (-3.27 pp).
- With 551 test images, gaps below ≈ 3.1 pp are within noise (Section 22). Only A5 falls outside this band.
- More complexity did not help: A5 (1.44 M extra parameters) was below each of its components.
- The modules that change the pretrained conv4_x→conv5_x features least stayed closest to the baseline: self-attention starts as identity ($\gamma = 0$), and spatial attention rescales only a 14×14 map.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

Channel Attention → Spatial Attention

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

How does the location of attention influence performance?

Results

Table 7: Experiment 2 – position of the Channel→Spatial module (ΔAcc relative to the baseline, 0.9437).

Model	Accuracy	Precision	Recall	Macro F1	ΔAcc (pp)	Attention params	Train time (min)	Inference (ms)
B1: after conv2_x 56×56×256	0.9093	0.9141	0.9068	0.9079	−3.45	8,563	30.4	16.85
B2: after conv3_x 28×28×512	**0.9510**	0.9539	0.9488	**0.9510**	+0.73	33,411	25.6	16.57
B3 (= A3): after conv4_x 14×14×1024	0.9238	0.9286	0.9212	0.9238	−2.00	132,259	20.6	18.56
B4: after conv5_x 7×7×2048	0.9310	0.9328	0.9309	0.9317	−1.27	526,563	19.3	16.67

Observations and inference

- **Intermediate features (P_2) were best:** B2 scored 0.9510, the highest of the study (+0.73 pp, within noise).
- **High-resolution low-level features (P_1) were worst:** 0.9093 (-3.45 pp, beyond noise). Edge-level channels carry little class information, and the change propagates through three more pretrained stages.
- **P_3 and P_4 lay in between.** P_4 needs the most parameters (527 k) but its 7×7 map leaves little room for spatial selection.
- Training time grows the earlier the module is inserted (30.4 min at P_1 → 19.3 min at P_4), because gradients must pass through all later stages.
- Position alone changed accuracy by 4.2 pp with the same module.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.

Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

Results

Table 8: Experiment 3 – attention at one or several positions (ΔAcc relative to the baseline).

Model	Accuracy	Precision	Recall	Macro F1	ΔAcc (pp)	Attention params	Train time (min)	Inference (ms)
Only at P_3 (= A3)	0.9238	0.9286	0.9212	0.9238	−2.00	132,259	20.6	18.56
Only at P_4 (= B4)	0.9310	0.9328	0.9309	0.9317	−1.27	526,563	19.3	16.67
$P_3 + P_4$ (C3)	**0.9383**	0.9374	0.9392	**0.9382**	−0.54	658,822	22.0	17.48
$P_2 + P_3 + P_4$ (C4)	0.8947	0.8975	0.8938	0.8946	−4.90	692,233	27.2	17.69

Observations and inference

- **No.** $P_3 + P_4$ (0.9383) was slightly better than either position alone but still below the baseline. $P_2 + P_3 + P_4$ gave the **worst result of the study** (0.8947, −4.90 pp), even though P_2 alone was the best single position.
- The three-stage model adds 692 k parameters and 6.6 min of training compared with P_3 alone, so the extra cost is not justified. Stacked modules compound the shifts in the features passed to the pretrained stages.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

Results

Table 9: Experiment 4 – order of channel and spatial attention at P_3.

Model	Accuracy	Precision	Recall	Macro F1	ΔAcc (pp)	Test loss	Best val. loss (epoch)	Attention params	Inference (ms)
Channel → Spatial $F \to M_c \to M_s$ (A3)	0.9238	0.9286	0.9212	0.9238	−2.00	0.2060	0.2402 (13)	132,259	18.56
Spatial → Channel $F \to M_s \to M_c$ (D1)	**0.9328**	0.9372	0.9314	**0.9335**	−1.09	0.1886	0.2818 (10)	132,259	17.72

Observations and inference

With identical parameters, Spatial→Channel was better on every metric (0.9328 versus 0.9238, test loss 0.1886 versus 0.2060), but the 0.91 pp gap is within noise. Spatial-first computes its map from unscaled features, before the channel gates switch off about half of the channels (Section 28.1). It also reached its best validation loss earlier (epoch 10 versus 13).

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	0.9056	0.9040	18.46 ms
5×5	0.9165	0.9162	17.30 ms
7×7	0.9238	0.9238	18.56 ms

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

Observations and inference

Inside the Channel→Spatial module at P_3, accuracy rose **monotonically** with k (0.9056 → 0.9165 → 0.9238, about +1.8 pp). On a 14×14 map a 7×7 kernel sees half the map, so a larger kernel gives useful context at negligible cost: +80 parameters and no measurable change in inference time. No kernel size lifted the module above the baseline.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

Mechanisms selected

All five mechanisms were evaluated at P_3:

- **SE** (= Channel, A1);
- **ECA**, a 1-D convolution with $k = 5$ over the channel descriptor (5 parameters, F1);
- **CBAM**, an avg+max shared MLP followed by 7×7 spatial attention (132,259 parameters, F2);
- **Coordinate Attention**, which pools along H and W separately (100,512 parameters, F3, the "Additional Mechanism");
- **Self-Attention** (A4).

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	0.9437	0.9489	0.9409	0.9436	23,597,957 (+0)	12.85 ms
Channel / SE	0.9183	0.9218	0.9178	0.9192	23,730,117 (+132,160)	18.86 ms
Spatial Attention	0.9347	0.9379	0.9312	0.9340	23,598,056 (+99)	15.78 ms
ECA	0.9474	0.9498	0.9487	0.9485	23,597,962 (+5)	19.06 ms
CBAM	0.9220	0.9257	0.9189	0.9214	23,730,216 (+132,259)	18.37 ms
Self-Attention	0.9419	0.9438	0.9392	0.9411	24,908,678 (+1,310,721)	17.03 ms
Additional Mechanism (Coordinate Attention)	0.9129	0.9202	0.9103	0.9135	23,698,469 (+100,512)	18.88 ms

Parameters = total (attention module in parentheses); *Inference Time* per image.

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance + Complexity + Attention Behaviour}}$$

rather than considering only classification accuracy.

Answers

1. Accuracy ranged from 0.9111 (Combined) to 0.9474 (ECA). Only ECA exceeded the baseline (+0.36 pp, within noise).
2. Macro F1 tracked accuracy almost exactly ($|F1 - \text{Acc}| \leq 0.0011$) with the same ranking. No mechanism changed class balance.
3. ECA 5; Spatial 99; Coordinate 100,512; SE 132,160; CBAM 132,259; Self-Attention 1,310,721 (up to 5.6 % of the model).
4. Analytically only self-attention adds real computation (+7.4 %, Table 10); the others add less than 0.06 %. Measured times (15.8–19.1 ms versus 12.85 ms) were inflated by heat on the fan-less MacBook Air: the baseline ran first on a cool chip, and ECA was measured as "slowest" with only 5 parameters.

Table 10: Analytical extra computation of each module (millions of multiply–accumulate operations per image; ResNet50 $\approx$ 4.09 GMAC). Pooling and element-wise rescaling are included.

Module	P_1	P_2	P_3	P_4	% of ResNet50 at P_3
Channel / SE	0.81	0.43	0.33	0.62	0.008 %
Spatial (7×7)	2.72	1.28	0.62	0.31	0.015 %
Channel→Spatial	3.53	1.72	0.95	0.93	0.023 %
CBAM	3.53	1.75	1.08	1.45	0.027 %
ECA	0.80	0.40	0.21	0.11	0.005 %
Coordinate	2.06	1.72	2.24	3.87	0.055 %
Self-attention	3,090.04	611.34	301.36	262.53	7.368 %

5. Partly. Most attention models reached 90 % validation accuracy earlier (epoch 2–6 versus 8), but none reached a lower validation loss than the baseline, and all had a larger train–validation gap (Section 24).
6. Spatial-type modules gave slightly more compact Grad-CAMs (coverage 0.18–0.20 versus 0.22). Self-attention and Coordinate attention gave broader ones (0.28 and 0.37, Table 14).
7. No. The most complex modules (Combined, Self-Attention) did not give the best accuracy.
8. Yes. ECA (5 parameters) and Spatial (99) matched or beat Self-Attention (1.31 M), CBAM and SE (132 k).

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

Strategy used (identical for every model)

Table 11: Transfer-learning protocol ("protocol card") used for every experiment.

Setting	Value (identical for all models)
Backbone	Keras ResNet50, ImageNet weights, `include_top=False`, input $224 \times 224 \times 3$
Classifier head	GlobalAveragePooling2D → Dropout(0.3) → Dense(5) → softmax
Phase 1 (steps 7–8)	all ResNet50 layers frozen; attention + head trained for 10 epochs, Adam, LR 10^{-3}
Phase 2 (steps 9–10)	conv5_x unfrozen (BatchNorm kept frozen), 10 epochs, Adam, LR 10^{-4}
LR schedule	ReduceLROnPlateau on validation loss, factor 0.5, patience 2, min. LR 10^{-7} (in each phase)
Batch size / seed	32 / 42
Model selection	no early stopping (same 20 epochs for all); weights of the epoch with the lowest validation loss restored
Loss	categorical cross-entropy
Trainable parameters	baseline: 10,245 (Phase 1), 14,963,717 (Phase 2); attention models: + attention parameters

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

Implementation

One-hot labels, a float32 softmax output and Keras `categorical_crossentropy` were used for every model.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\,\frac{\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

How the metrics were computed

- Metrics were computed with scikit-learn (`average="macro"`) on the 551 test images, using the best-validation-loss weights.
- Inference time is the median of 3 timed passes over the test set (batch 32, after warm-up), reported in ms per image.
- **Noise level:** with $n = 551$ test images the 95 % confidence interval of an accuracy is $\approx \pm 2.2$ pp. Gaps below ≈ 3.1 pp between two models are therefore not meaningful.

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9437	0.9489	0.9409	0.9436	23,597,957 (+0)	12.85 ms
Channel Attention	P_3	0.9183	0.9218	0.9178	0.9192	23,730,117 (+132,160)	18.86 ms
Spatial Attention	P_3	0.9347	0.9379	0.9312	0.9340	23,598,056 (+99)	15.78 ms
Channel+Spatial	P_2	0.9510	0.9539	0.9488	0.9510	23,631,368 (+33,411)	16.57 ms
Channel+Spatial	P_3	0.9238	0.9286	0.9212	0.9238	23,730,216 (+132,259)	18.56 ms
Channel+Spatial	P_4	0.9310	0.9328	0.9309	0.9317	24,124,520 (+526,563)	16.67 ms
Self-Attention	P_3	0.9419	0.9438	0.9392	0.9411	24,908,678 (+1,310,721)	17.03 ms
Additional Attention-1	P_3	0.9474	0.9498	0.9487	0.9485	23,597,962 (+5)	19.06 ms
Additional Attention-2	P_3	0.9220	0.9257	0.9189	0.9214	23,730,216 (+132,259)	18.37 ms

Parameters = total (attention module in parentheses). Additional Attention-1 = ECA, Additional Attention-2 = CBAM.

Observations

Channel+Spatial@P_2 (0.9510), ECA (0.9474), the baseline (0.9437) and Self-Attention (0.9419) are the top group and statistically indistinguishable. Channel (0.9183) and CBAM (0.9220) are the lowest. No added parameters bought a significant gain.

24. Graphical Performance Analysis

Students should not present only numerical tables.

Appropriate graphs must be generated to analyze model behavior.

At least three of the following graphical comparisons should be included in the laboratory report.

Graphs included

All six comparisons (24.1–24.6) are included.

24.1 Training and Validation Accuracy

Plot

Epoch versus Accuracy

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

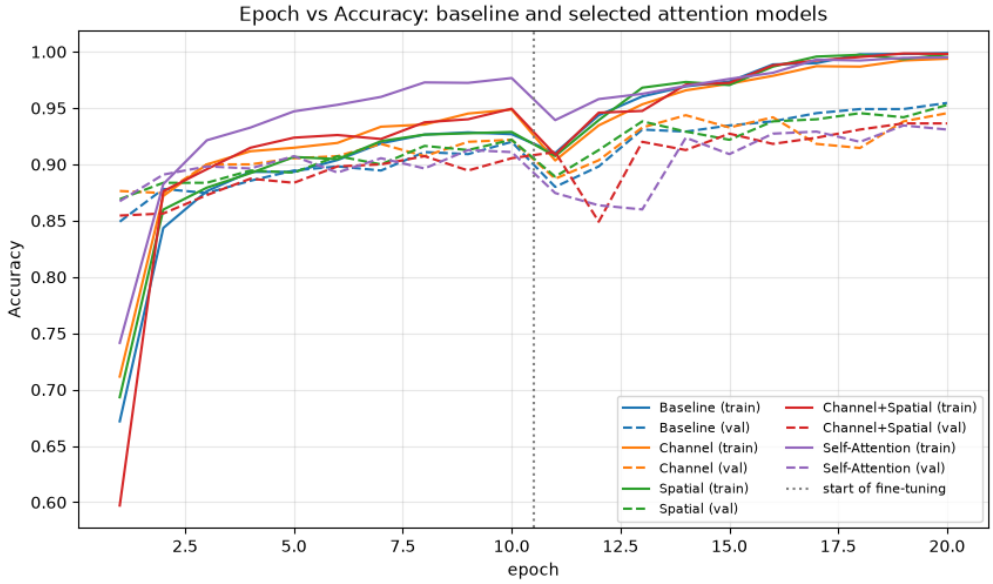


Figure 3: Training (solid) and validation (dashed) accuracy. The dotted line marks the start of fine-tuning.

Table 12: Convergence statistics of all 17 trained models (from the per-epoch training logs; epochs 1–10 = Phase 1, 11–20 = Phase 2).

Model	Pos.	Epoch val. acc $\geq$ 0.90	Min. val. loss	Kept epoch	Final train acc	Final val. acc	Train–val gap	Val. acc std (last 5)
A0 Baseline	–	8	0.2212	13	0.9988	0.9545	0.0443	0.0054
A1 Channel (= SE)	P_3	3	0.2747	6	0.9938	0.9455	0.0483	0.0127
A2 Spatial	P_3	6	0.2263	13	0.9977	0.9527	0.0450	0.0051
A3 Channel→Spatial	P_3	7	0.2402	13	0.9981	0.9364	0.0617	0.0072
A4 Self-Attention	P_3	5	0.2386	14	0.9949	0.9309	0.0640	0.0048
A5 Combined (C→S→SA)	P_3	2	0.2426	12	0.9946	0.9327	0.0619	0.0055
B1 Ch→Sp	P_1	12	0.3075	12	0.9988	0.9345	0.0643	0.0014
B2 Ch→Sp	P_2	6	0.2403	19	0.9996	0.9418	0.0578	0.0093
B4 Ch→Sp	P_4	3	0.2333	14	0.9992	0.9418	0.0574	0.0114
C3 Ch→Sp	P_3+P_4	5	0.2480	13	0.9977	0.9418	0.0559	0.0051
C4 Ch→Sp	P_2+P_3+P_4	6	0.3088	7	0.9977	0.9309	0.0668	0.0027
D1 Sp→Ch	P_3	9	0.2818	10	0.9961	0.9327	0.0634	0.0071
E1 Ch→Sp, $k = 3$	P_3	7	0.2823	12	0.9930	0.9218	0.0712	0.0058
E2 Ch→Sp, $k = 5$	P_3	4	0.2606	12	0.9981	0.9382	0.0599	0.0041
F1 ECA	P_3	10	0.2441	15	0.9984	0.9491	0.0493	0.0086
F2 CBAM	P_3	12	0.2635	14	0.9973	0.9418	0.0555	0.0078
F3 Coordinate	P_3	3	0.2906	6	0.9852	0.9309	0.0543	0.0024

Comments

- **Convergence:** faster only early on. Most attention models crossed 90 % validation accuracy before the baseline (epoch 8), but none ended better.
- **Smoothness:** spatial and self-attention were as smooth as the baseline (validation-accuracy std $\approx$ 0.005). Channel attention was the most jumpy (0.013).
- **Train–validation gap:** not reduced. The baseline had the smallest gap (0.044); the attention models had 0.045–0.071.
- **Overfitting:** all models reach $\approx$1.00 training accuracy in Phase 2 while validation accuracy saturates at 0.93–0.95. The best-validation-loss weights were restored.
- **Instability:** every model dipped briefly when conv5_x was unfrozen (epoch 11), deepest for channel+spatial (validation accuracy 0.849) and self-attention (0.860). All recovered by epoch 13–14.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models.

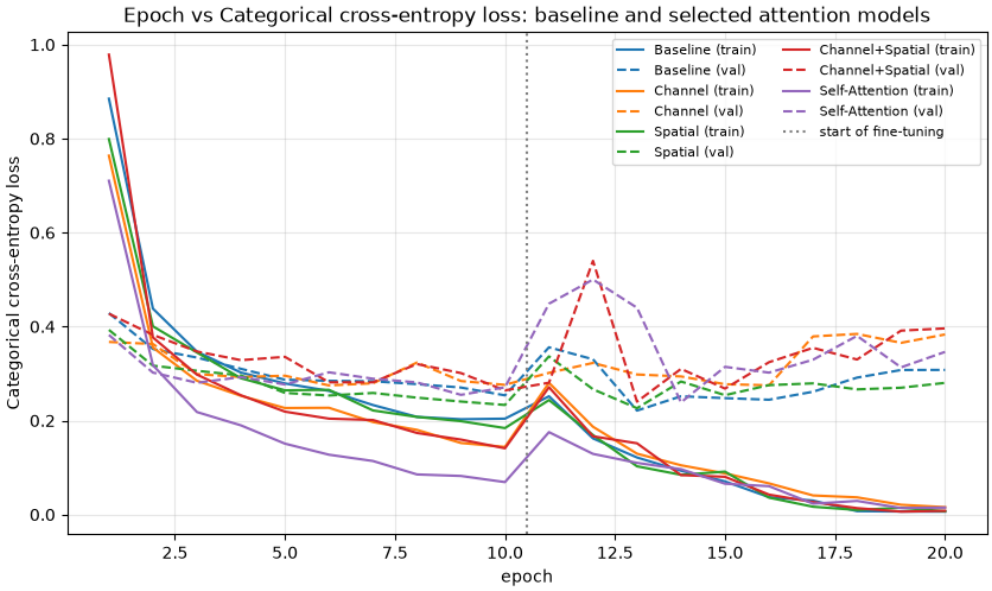


Figure 4: Training (solid) and validation (dashed) categorical cross-entropy loss.

Comparison of convergence

Self-attention reached the lowest Phase 1 training loss (0.069), but not a lower validation loss. The baseline reached the **lowest validation loss of all 17 models** (0.2212, epoch 13). After epochs 13–15 the validation loss of every model rises again (e.g. channel+spatial 0.40 at epoch 20) while the training loss approaches 0, which is clear over-fitting. Attention models fit the training set as fast as the baseline but do not converge to a better validation optimum.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.

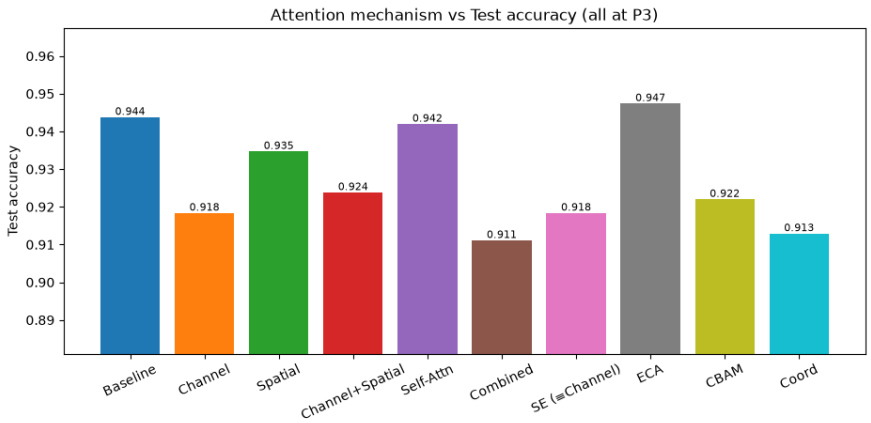


Figure 5: Test accuracy per mechanism at P_3 (y-axis starts at 0.88; SE ≡ Channel).

Observation

Only ECA (0.947) is above the baseline (0.944). All the others except Combined (0.911) lie within 3 pp of the baseline.

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

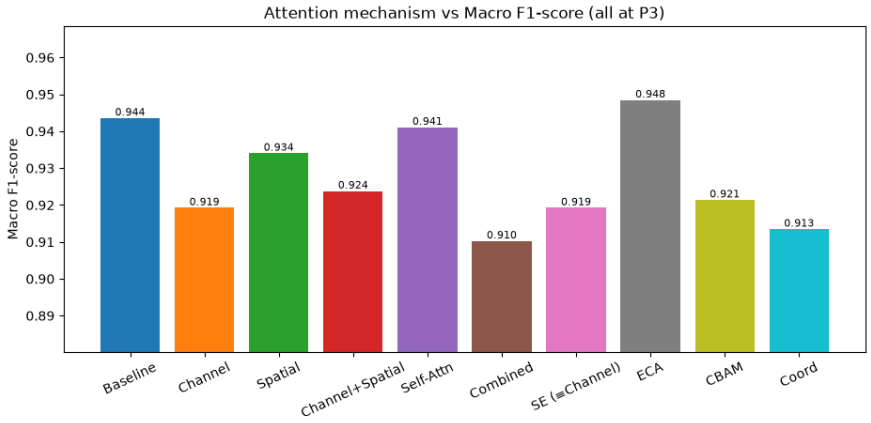


Figure 6: Macro F1-score per mechanism at P_3.

Observation

The macro-F1 bars mirror the accuracy bars, so no mechanism traded overall accuracy for class balance.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

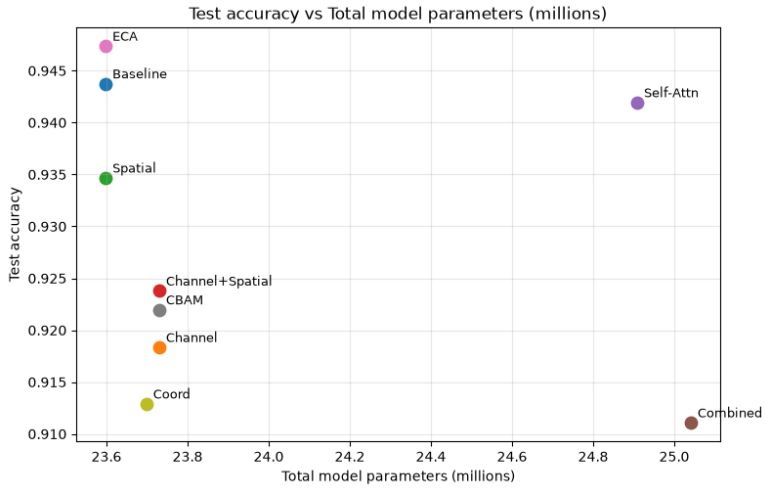


Figure 7: Test accuracy versus total parameters.

Observation

Size and accuracy are not positively related (correlation -0.18). The cheapest models (Baseline, ECA, Spatial) are the most accurate, and the efficient frontier is the baseline and ECA alone.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

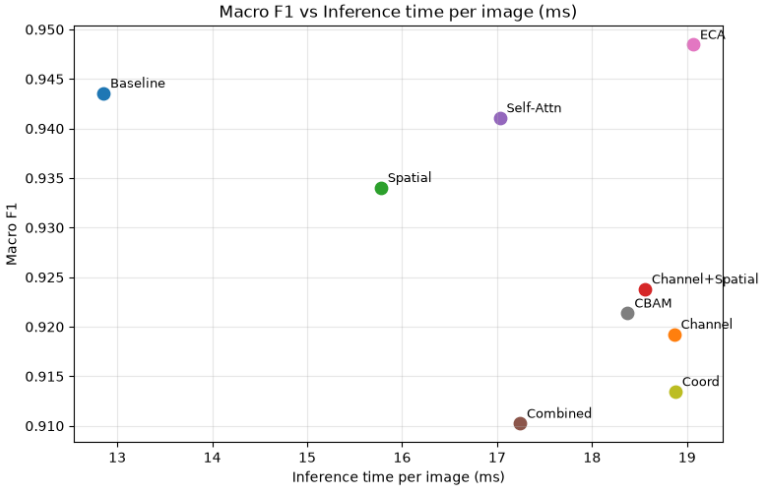


Figure 8: Macro F1 versus measured inference time per image.

Observation

No. The baseline is the fastest model and among the best. The extra 3–6 ms of the attention models mostly reflects thermal throttling (Section 19.2, question 4), and the only module with a real extra cost (self-attention, +7.4 %) gave no F1 gain.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

Confusion matrices

The selected attention model is Channel+Spatial at P_3 (A3), the configuration shared by Experiments 1–5.

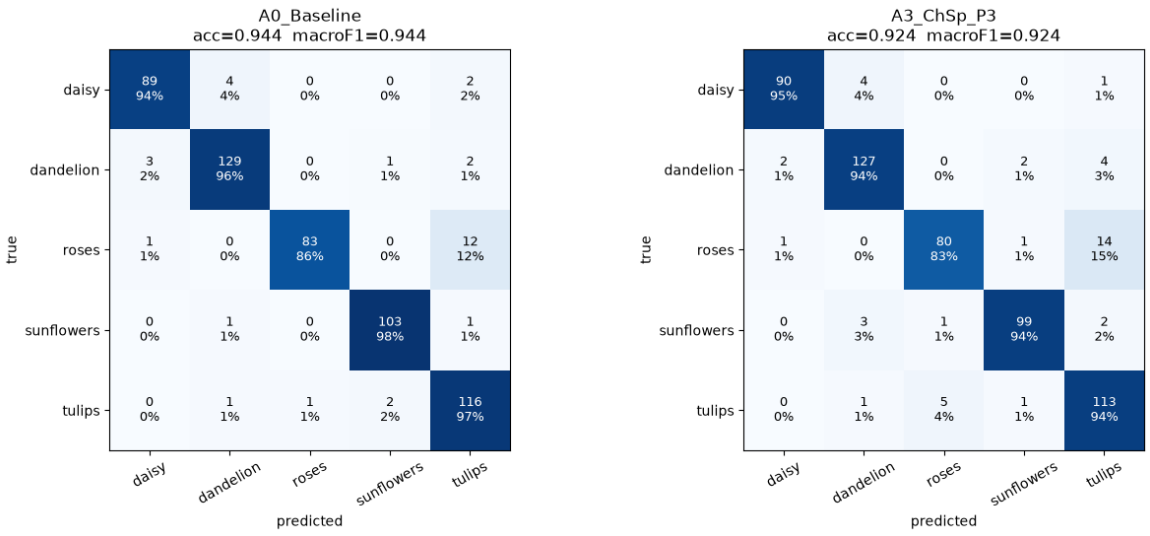


Figure 9: Confusion matrices (count and row %): baseline (left) and Channel+Spatial@P_3 (right).

Table 13: Per-class recall and F1 of the baseline and the selected attention model.

Class	Test images	Recall baseline	Recall Ch+Sp@P_3	ΔRecall	F1 baseline	F1 Ch+Sp@P_3	ΔF1
daisy	95	0.937	0.947	+0.011	0.947	0.957	+0.011
dandelion	135	0.956	0.941	-0.015	0.956	0.941	-0.015
roses	96	0.865	0.833	-0.031	0.922	0.879	-0.043
sunflowers	105	0.981	0.943	-0.038	0.976	0.952	-0.024
tulips	120	0.967	0.942	-0.025	0.917	0.890	-0.027

Analysis

- **Most accurate:** sunflowers (recall 0.981), tulips (0.967) and dandelion (0.956). Roses are the hardest class (0.865).
- **Most confused:** roses→tulips (12 baseline, 14 attention), then daisy→dandelion (4 each). These pairs share colour or radial structure.
- **Benefits most:** only daisy improved (F1 +0.011). Roses lost the most (F1 −0.043).
- **Confusion reduced?** No. Roses↔tulips grew from 13 to 19 errors.
- **Distributed?** Overall accuracy fell (509 versus 520 correct), and the losses were spread over four classes.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

Images used and intermediate feature maps

The same images were used in all qualitative figures. Test images 13 (dandelion), 440 (tulips) and 458 (daisy) are correct for all five main models. Test images 222 (roses → tulips) and 530 (dandelion → tulips) are errors of the selected model; the baseline also misclassifies 222. Figure 10 shows the features before and after the P_3 module (required visualizations 5 and 10).

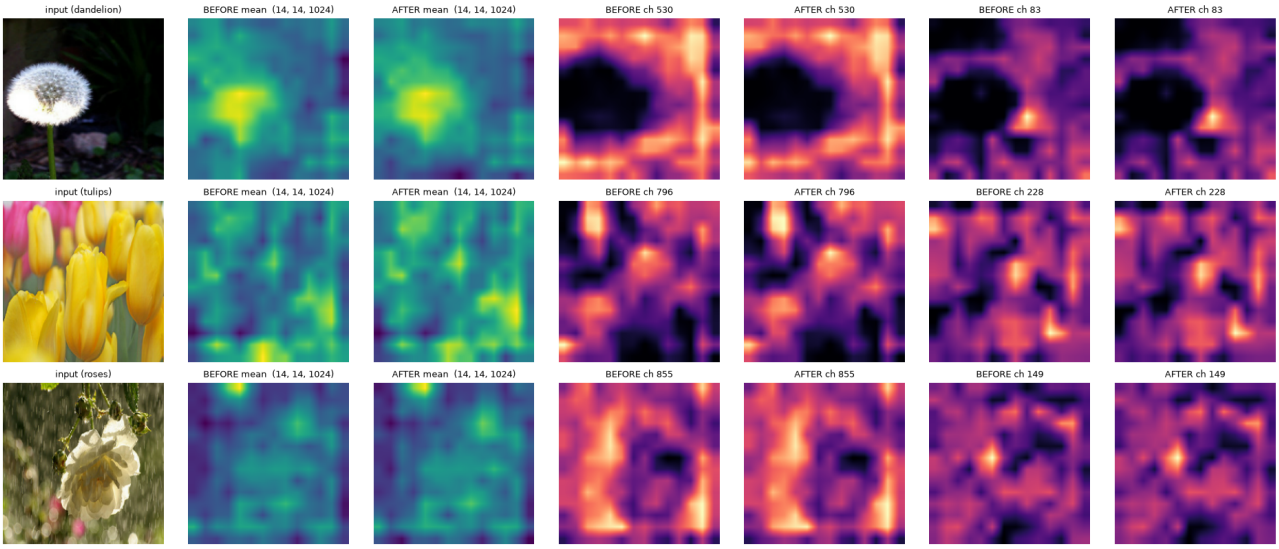


Figure 10: Feature maps before and after the Channel+Spatial module at P_3 (shape $14 \times 14 \times 1024$ in both cases).

Observation

The shape is unchanged and the spatial patterns are almost identical before and after attention. The module mainly rescales channels, with slight contrast changes near the bottom rows.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

Implementation

A^k is the $7 \times 7 \times 2048$ tensor feeding GAP, and y^c is the pre-softmax logit of the predicted class. The map is normalised, resized to 224×224 and overlaid (red = high).

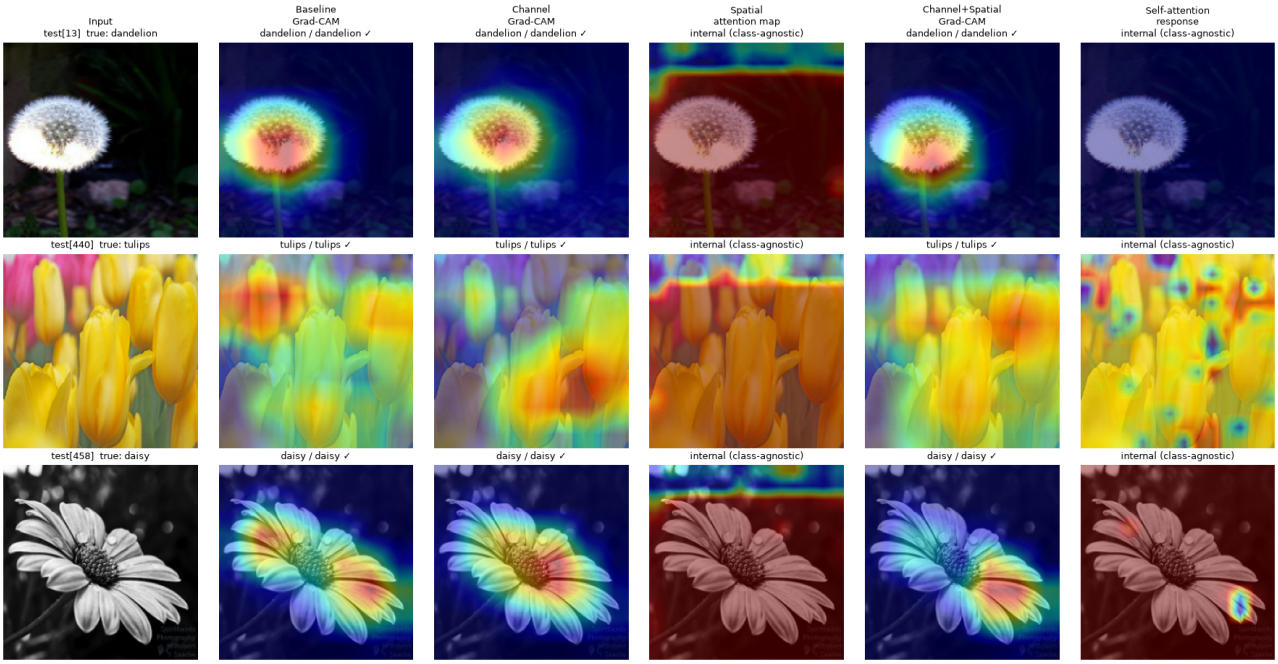


Figure 11: Input | Baseline Grad-CAM | Channel Grad-CAM | Spatial map (internal) | Channel+Spatial Grad-CAM | Self-attention response (internal) for three correctly classified images (true / predicted).

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

Interpretation (visible evidence)

- **Dandelion (13):** all models peak on the seed head with a dark background. The channel model spreads over the whole head, and channel+spatial concentrates on its lower-left half. Focus on the flower is about equal to the baseline.
- **Tulips (440):** activation is distributed across several tulip heads in all models. Only the peak moves: upper-left for the baseline, right for channel, centre-right for channel+spatial.
- **Daisy (458):** activation lies on the *petals*, not on the central disc. The baseline uses both sides, and channel+spatial mainly the right-hand petals.
- **Over 60 images** (Table 14): spatial-type models are slightly more compact than the baseline, with the same low border share. Self-attention and Coordinate attention spread more widely, i.e. less background suppression.
- **Errors** (Section 30): the focus moves to background objects.

Table 14: Grad-CAM statistics over the same 60 random test images. *Coverage* = fraction of pixels with normalised activation > 0.5 (smaller = more compact). *Border share* = fraction of the total activation lying in the outer frame of width 28 pixels (12.5 % of the side); lower means more central.

Model (attention at P_3)	**Activation coverage** (mean ± std)	**Border share** (mean ± std)
Baseline	0.220 ± 0.104	0.193 ± 0.117
Channel	0.213 ± 0.125	0.233 ± 0.142
Spatial	0.178 ± 0.104	0.195 ± 0.132
Channel+Spatial	0.199 ± 0.126	0.192 ± 0.114
Self-Attention	0.282 ± 0.159	0.263 ± 0.113
ECA	0.227 ± 0.129	0.209 ± 0.122
CBAM	0.185 ± 0.103	0.207 ± 0.145
Coordinate	0.372 ± 0.163	0.232 ± 0.085
Uniform map (reference)	–	0.438

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

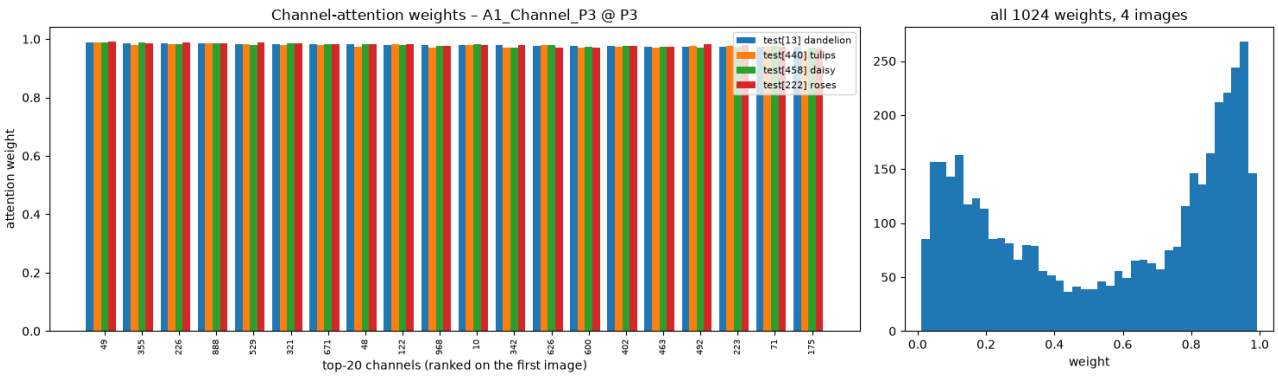


Figure 12: Channel weights of A1 at P_3: top-20 channels for four images (left) and all 4×1024 weights (right).

Comments

- **A few strong channels?** No. The top-20 (weights ≈0.98) hold only 3.5 % of the total weight; uniform weighting would give 2.0 %.
- **Many similar channels?** Yes. The weights are **bimodal**: many channels near 1, many near 0 (56 % above 0.5). The module acts as a soft channel selector.
- **Change between images?** Hardly. The weight vectors of different images correlate at 0.991–0.998. The module learned an almost *input-independent* gating, which explains its small effect.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

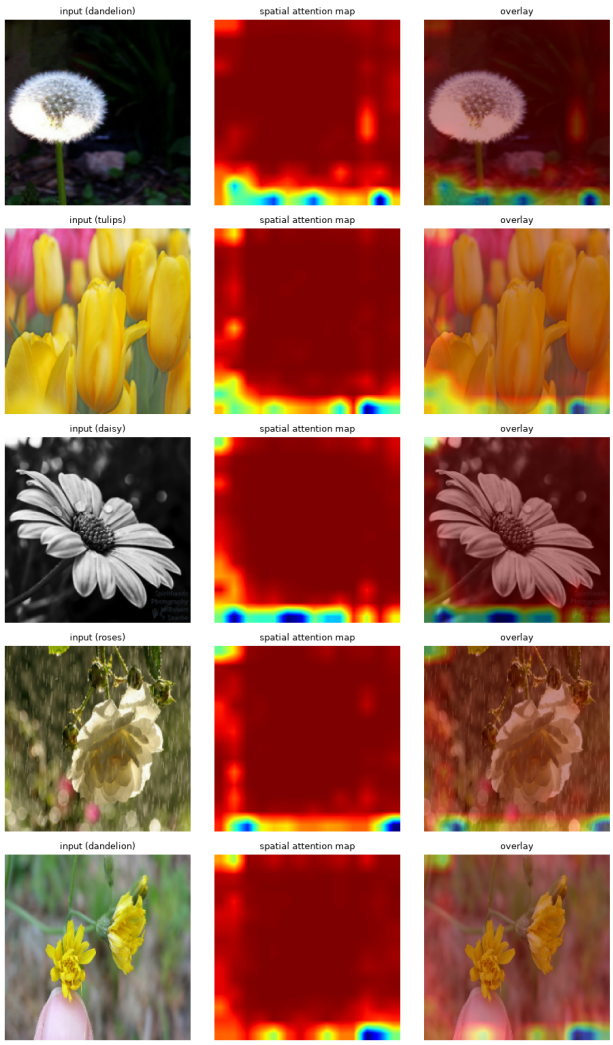


Figure 13: Input | Spatial-attention map | Overlay (Channel+Spatial, P_3; min–max normalised, red = high).

Regions receiving high attention

Attention is high over almost the **whole interior**, covering flower and background alike. It is low only along the **bottom border** and at a few edge spots. The maps are nearly identical for all images, so the module learned

a positional prior that down-weights the zero-padded border rather than a flower mask. The spatial-only model (A2) does the same at the top border.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution. Students should analyze how distant regions interact within the feature map.

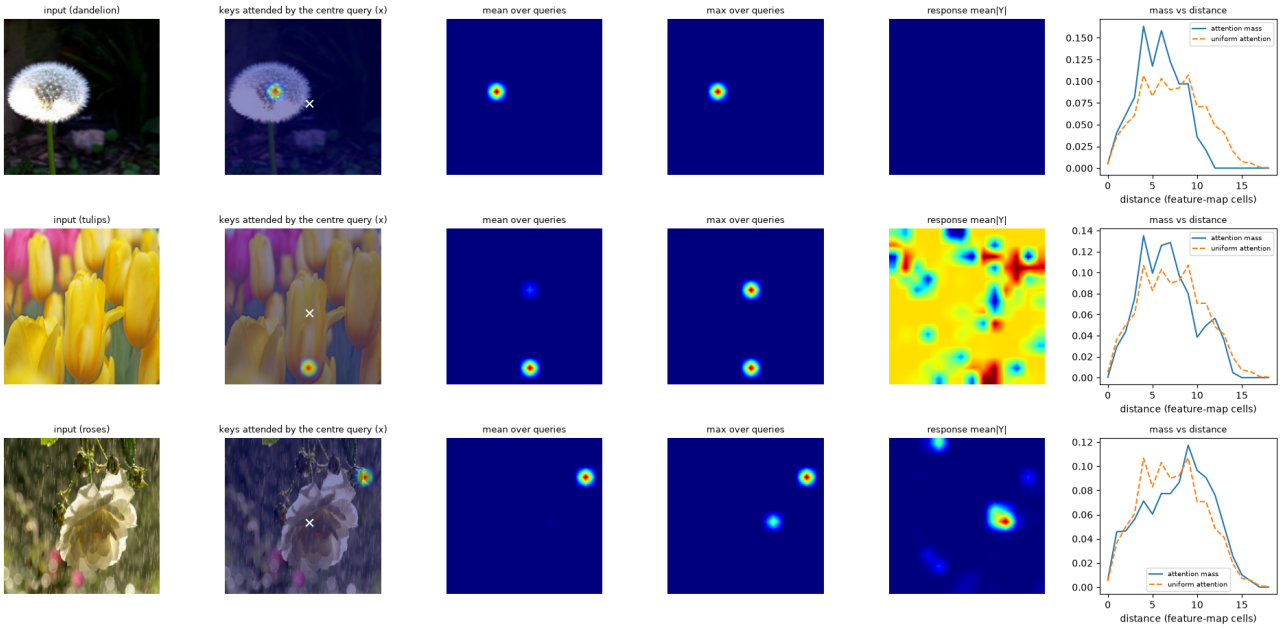


Figure 14: Self-attention (A4, P_3): keys of the centre query (x), mean and max over queries, response mean$|Y|$, and attention mass versus distance.

Analysis

- Practically all 196 queries attend to the *same one or two key positions*, so the mean and max maps show the same hot spot as the single query. The attention has collapsed to nearly identical rows. For the dandelion the response is even constant over the map.
- **Distant regions:** the attention mass follows the uniform curve up to ≈10 cells (≈160 pixels), so distant regions interact as strongly as near ones; there is no locality bias.
- With $\gamma = 0.066$, the module adds a small global-context vector to every position. Accuracy therefore stays near the baseline while Grad-CAM becomes broader.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.

Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.

Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;

- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

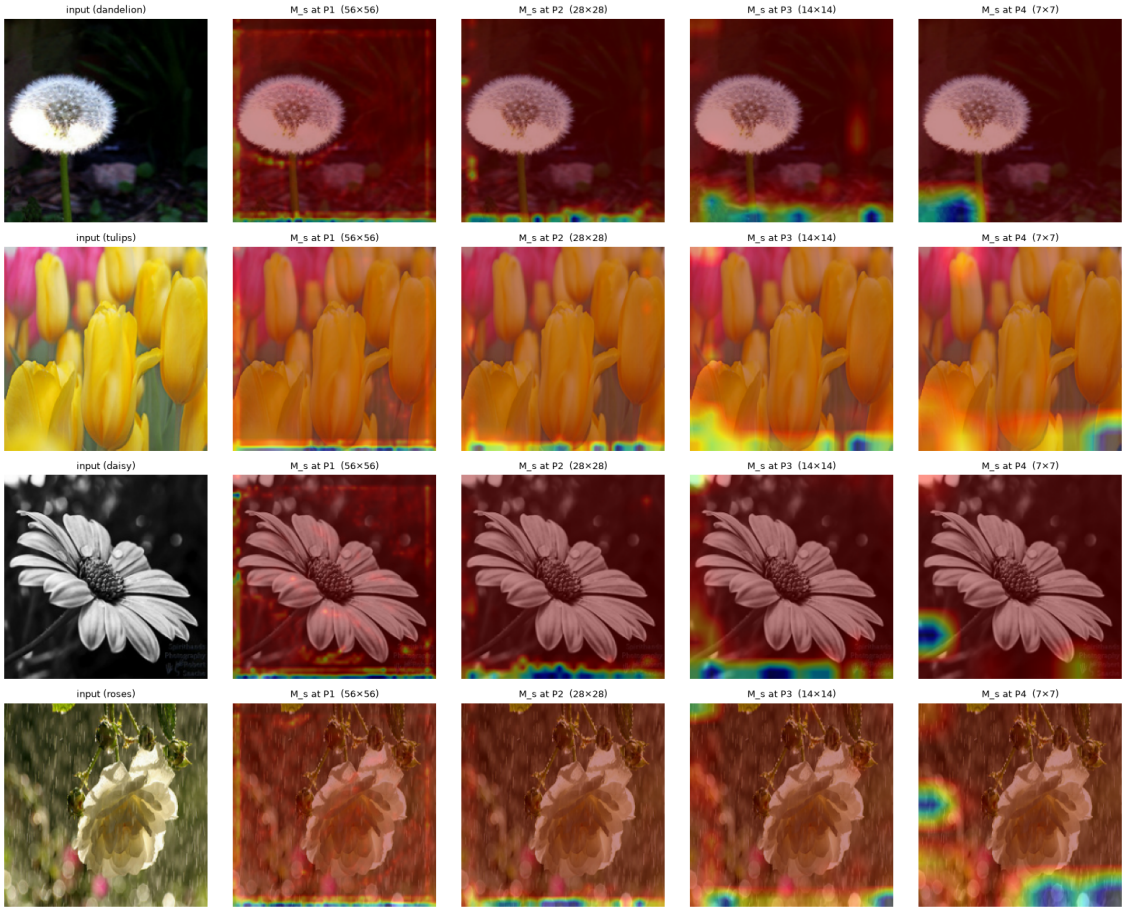


Figure 15: Input | $M_s^{P_1}$ | $M_s^{P_2}$ | $M_s^{P_3}$ | $M_s^{P_4}$ (models B1, B2, A3, B4) on the same images.

Observations

- P_1/P_2**:** fine-grained and nearly uniform maps with low values along the border. Only P_1 shows faint edge texture (daisy petals, dandelion outline).
- P_3**:** a thicker low band at the bottom plus a few edge spots; no flower parts are singled out.
- P_4**:** coarse 32×32-pixel blocks. Whole corner regions are switched off (e.g. lower-left for the dandelion).
- Deeper maps become coarser and more region-like, but at no depth does the map isolate the flower. The sheet's hypotheses are only partly confirmed.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.

A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

Selected images

The 3 correct images are in Figure 11; the 2 incorrect ones are below (arranged horizontally).

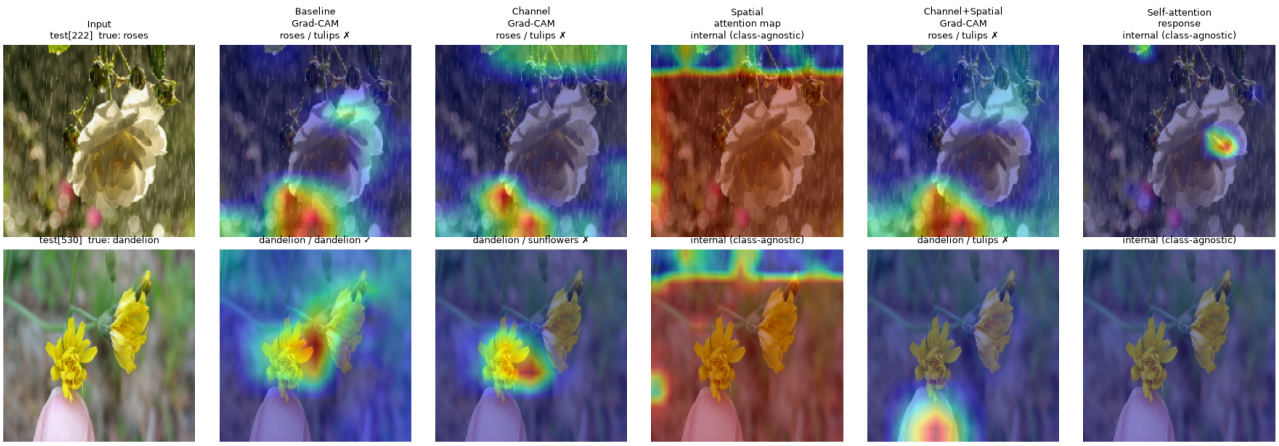


Figure 16: Same layout as Figure 11 for two incorrectly classified images (✗ = wrong).

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

Discussion of the criteria

- **Foreground, petals, centre:** for the correct images every Grad-CAM peaks on the flower: the seed head, the tulip heads, and the daisy *petals* rather than its dark centre.
- **Background suppression:** dark and blurred backgrounds stay low in all models. The internal spatial maps suppress only the image border.
- **Compactness / coverage:** spatial-type attention is slightly more compact (coverage 0.18–0.20 versus 0.22); self-attention (0.28) and Coordinate attention (0.37) are broader.
- **Incorrectly attended regions:** in image 222 (a white rose) the baseline, channel and channel+spatial models predict tulips from the *pink background blobs*. In image 530 channel+spatial predicts tulips from a *pink fingertip*, whereas the correct baseline looks at the flower heads.
- **Correct versus incorrect:** correct predictions rely on the flower; errors rely on background colour cues.
- **Shallow versus deep:** fine border and edge patterns at P_1/P_2, coarse regions at P_4 (Section 29).
- **Consistency:** the internal maps are nearly identical across images, whereas Grad-CAM varies with image and class.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

Where each visualization is

1: Fig. 1; 2: Fig. 3; 3: Fig. 4; 4: Fig. 9; 5 and 10: Fig. 10; 6: Fig. 12; 7 and 8: Fig. 13; 9: Fig. 14; 11 and 12: Figs. 11, 16, 17; 13: Fig. 15; 14: Figs. 5–8.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input | Baseline | SE | ECA | CBAM | Self-Attention

using appropriate visualization methods.

Figures produced

The recommended layout is used in Figures 11 and 16. The extended layout, with Coordinate attention added, is below.

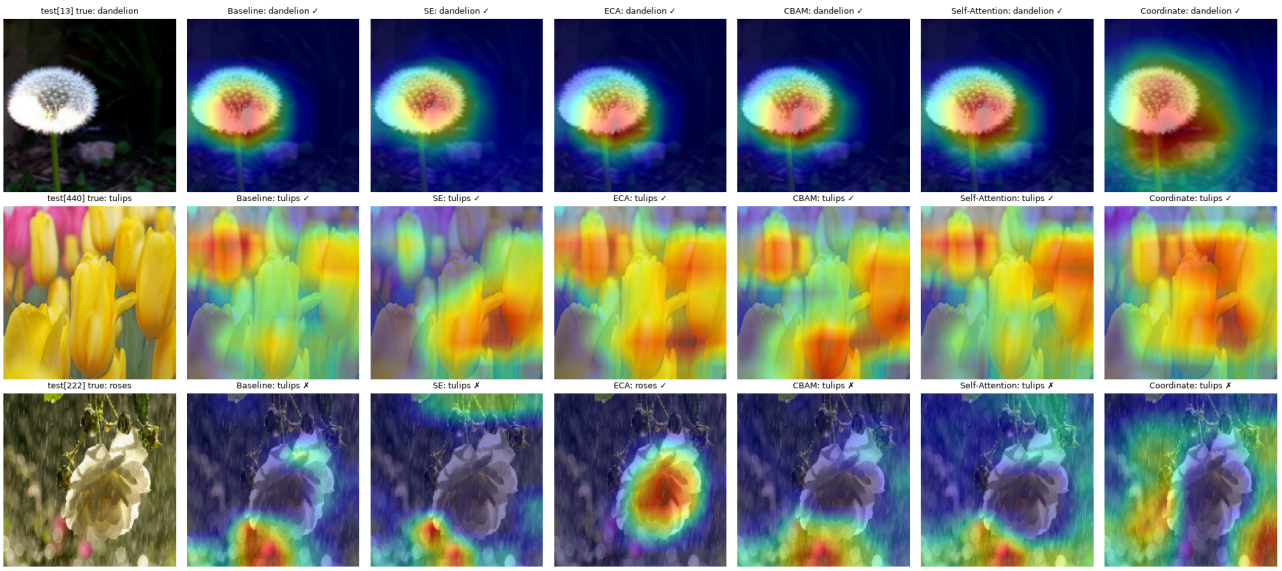


Figure 17: Input | Baseline | SE | ECA | CBAM | Self-Attention | Coordinate (Grad-CAM, P_3).

Observation

For the white rose (222) only **ECA** predicts roses, and its map is centred on the rose. All other models, including the baseline, predict tulips from the pink background. Coordinate attention also spreads into the background for the dandelion, matching its high coverage.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

Table 15: Quantitative results of all 17 trained configurations (test set). Differences below ≈ 3.1 pp are within the sampling noise.

Model	Position	Accuracy	Correct / 551	Precision	Recall	Macro F1	ΔAcc (pp)
A0 Baseline	–	0.9437	520	0.9489	0.9409	0.9436	+0.00
A1 Channel (= SE)	P_3	0.9183	506	0.9218	0.9178	0.9192	−2.54
A2 Spatial	P_3	0.9347	515	0.9379	0.9312	0.9340	−0.91
A3 Channel→Spatial	P_3	0.9238	509	0.9286	0.9212	0.9238	−2.00
A4 Self-Attention	P_3	0.9419	519	0.9438	0.9392	0.9411	−0.18
A5 Combined (C→S→SA)	P_3	0.9111	502	0.9120	0.9104	0.9102	−3.27
B1 Ch→Sp	P_1	0.9093	501	0.9141	0.9068	0.9079	−3.45
B2 Ch→Sp	P_2	0.9510	524	0.9539	0.9488	0.9510	+0.73
B4 Ch→Sp	P_4	0.9310	513	0.9328	0.9309	0.9317	−1.27
C3 Ch→Sp	$P_3{+}P_4$	0.9383	517	0.9374	0.9392	0.9382	−0.54
C4 Ch→Sp	$P_2{+}P_3{+}P_4$	0.8947	493	0.8975	0.8938	0.8946	−4.90
D1 Sp→Ch	P_3	0.9328	514	0.9372	0.9314	0.9335	−1.09
E1 Ch→Sp, $k = 3$	P_3	0.9056	499	0.9044	0.9074	0.9040	−3.81
E2 Ch→Sp, $k = 5$	P_3	0.9165	505	0.9171	0.9185	0.9162	−2.72
F1 ECA	P_3	0.9474	522	0.9498	0.9487	0.9485	+0.36
F2 CBAM	P_3	0.9220	508	0.9257	0.9189	0.9214	−2.18
F3 Coordinate	P_3	0.9129	503	0.9202	0.9103	0.9135	−3.09

Inference

Only B2 (+0.73 pp) and ECA (+0.36 pp) scored above the baseline, both within noise. Four configurations were clearly worse (> 3.1 pp): $P_2 + P_3 + P_4$, $k = 3$, P_1 and Combined. Precision, recall and F1 move together for every model.

33.2 Complexity Analysis

Report:

Parameters + Inference Time

Table 16: Complexity of all 17 trained configurations. Extra MACs are analytical; train and inference times were measured on the fan-less MacBook Air and include thermal-throttling effects.

Model	Position	Total params	Trainable (Phase 2)	Attention params	Extra MACs (M / image)	Train time (min)	Inference (ms / image)
A0 Baseline	–	23,597,957	14,963,717	0	0.00	13.1	12.85
A1 Channel (= SE)	P_3	23,730,117	15,095,877	132,160	0.33	20.6	18.86
A2 Spatial	P_3	23,598,056	14,963,816	99	0.62	20.3	15.78
A3 Channel→Spatial	P_3	23,730,216	15,095,976	132,259	0.95	20.6	18.56
A4 Self-Attention	P_3	24,908,678	16,274,438	1,310,721	301.36	22.0	17.03
A5 Combined (C→S→SA)	P_3	25,040,937	16,406,697	1,442,980	302.31	23.4	17.24
B1 Ch→Sp	P_1	23,606,520	14,972,280	8,563	3.53	30.4	16.85
B2 Ch→Sp	P_2	23,631,368	14,997,128	33,411	1.72	25.6	16.57
B4 Ch→Sp	P_4	24,124,520	15,490,280	526,563	0.93	19.3	16.67
C3 Ch→Sp	P_3+P_4	24,256,779	15,622,539	658,822	1.88	22.0	17.48
C4 Ch→Sp	P_2+P_3+P_4	24,290,190	15,655,950	692,233	3.60	27.2	17.69
D1 Sp→Ch	P_3	23,730,216	15,095,976	132,259	0.95	21.1	17.72
E1 Ch→Sp, $k = 3$	P_3	23,730,136	15,095,896	132,179	0.94	20.9	18.46
E2 Ch→Sp, $k = 5$	P_3	23,730,168	15,095,928	132,211	0.94	21.0	17.30
F1 ECA	P_3	23,597,962	14,963,722	5	0.21	20.3	19.06
F2 CBAM	P_3	23,730,216	15,095,976	132,259	1.08	22.1	18.37
F3 Coordinate	P_3	23,698,469	15,064,165	100,512	2.24	21.1	18.88

Inference

Extra parameters range from 5 (ECA) to 1.44 M (Combined, +6.1 %); channel-type modules scale with C^2/r. Only self-attention adds real computation (+7.4 % at P_3, +75 % if placed at P_1). Training time grows for earlier and stacked insertions. Measured inference times mainly reflect the laptop's thermal state; the baseline was measured first, on a cool chip.

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

Inference

Grad-CAM shows that every model relies on the flower when correct and on background colour cues when wrong. The internal maps are largely static: input-independent channel gating, border-suppressing spatial maps, and a collapsed self-attention with $\gamma = 0.066$. This explains why attention added little to the strong pretrained baseline, so the three analyses agree.

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

Only Attention Position Should Change

and when studying attention type:

Only Attention Type Should Change.

How the protocol was enforced

All settings were fixed in one configuration cell (Table 11); the saved split, seed 42 and the augmenter were reused by every run. Each experiment was defined only by {position: attention type}, and identical configurations (B3 = E3 = A3, SE = A1) were trained once. **Environment:** MacBook Air (M4), TensorFlow 2.18.1 with tensorflow-metal 1.2.0 (GPU results verified against CPU), Keras 3.8.0, Python 3.12. The only uncontrolled factor was chip temperature, which affects timings but not the accuracy metrics.

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

Completion status

All ten items were completed. Items 1–7 are models A0, A1, A2, A3, B2, B4 and A4 (Section 23); item 8 is Figures 11, 16 and 17; item 9 is Figure 15; item 10 is Figures 5–8. Ten further configurations were trained (A5, B1, C3, C4, D1, E1, E2, F1–F3). The optional self-attention at P_4 was not trained.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

Exploration carried out

All five mechanisms were integrated at P_3 under the same protocol. The quantitative comparison is in Section 19.1 and the qualitative comparison in Figure 17. ECA was the best model at P_3 (0.9474, 5 parameters). CBAM (0.9220) matched our Channel+Spatial module (0.9238), and Coordinate attention (0.9129) gave the broadest Grad-CAMs.

37. Discussion

1. What is the purpose of residual connections in ResNet50?
   Answer: A block computes $y = \mathcal{F}(x) + x$. The identity shortcut gives gradients a direct path, which prevents vanishing gradients and degradation, and each block only learns a correction to its input. This makes 50 layers trainable. Our self-attention uses the same idea ($F + \gamma Y$, $\gamma = 0$ at the start), which is why it stayed close to the baseline (0.9419).
2. What does conv2_x represent compared with conv5_x?
   Answer: conv2_x ($56 \times 56 \times 256$) has a small receptive field and encodes edges and textures; its maps still resemble the image (Figure 2). conv5_x ($7 \times 7 \times 2048$) has a very large receptive field and encodes class-discriminative semantics. It holds 63.6 % of the backbone parameters.
3. Why does the spatial resolution decrease with network depth?
   Answer: Stride-2 convolutions and pooling enlarge the receptive field, so local evidence can be aggregated into object-level evidence. Down-sampling also adds some translation invariance and keeps computation and memory affordable.
4. Why does the number of feature channels increase with network depth?
   Answer: Deeper layers represent many more abstract combinations (parts, configurations, objects), so they need more detectors. Doubling C while halving H and W keeps the computation per stage roughly balanced.
5. What is meant by low-level, intermediate, and high-level features?
   Answer: Low-level features are edges, colours and simple textures (stem, conv2_x). Intermediate features are local patterns and flower parts (conv3_x, conv4_x). High-level features are whole-object, class-specific semantics (conv5_x). Figure 2 shows this progression, from tulip outlines to tulip-head blobs to coarse evidence regions.
6. What is the purpose of an attention mechanism?
   Answer: To re-weight features depending on the input, amplifying informative channels or regions and suppressing irrelevant ones at low cost. Our results show that it only helps if the module actually learns image-specific weights. Ours mostly learned static weights, so the gains stayed within noise.
7. What does channel attention attempt to learn?
   Answer: Which channels (feature detectors) matter for the current image. Our module learned bimodal weights that were almost identical for every image (correlation ≥ 0.991, Figure 12), i.e. a fixed channel selection. This is why Channel/SE (0.9183) did not improve on the baseline.
8. What does spatial attention attempt to learn?
   Answer: Where the evidence is, as one weight per location. Our maps were high almost everywhere and low only along the image border, in all five test images (Figure 13). The module learned a positional prior rather than a flower mask, so spatial attention (0.9347) stayed near the baseline.
9. How is channel attention different from spatial attention?
   Answer: Channel attention produces a vector in $\mathbb{R}^C$ shared by all positions, by pooling over space and applying an MLP (132,160 parameters at P_3). Spatial attention produces a map in $\mathbb{R}^{H \times W}$ shared by all channels, by pooling over channels and applying a convolution (99 parameters). They act on orthogonal axes and differ in cost by more than 1000×.
10. Why can channel and spatial attention be used sequentially?
    Answer: They are complementary ("what" and "where"), and applying them in sequence factorises $H \cdot W \cdot C \approx 200\,\text{k}$ weights into $C + HW = 1{,}220$, which is far cheaper. In our runs the sequence (0.9238) did not beat spatial attention alone (0.9347). A plausible reason is that the channel gates first switch off about half of the channels from which the spatial map is then computed.
11. Why might channel attention behave differently at conv3_x and conv5_x?
    Answer: conv3_x channels are generic mid-level detectors, and re-weighting them affects two more pre-trained stages. conv5_x channels are semantic and feed the classifier directly, and the MLP there is 16× larger (527 k versus 33 k parameters for the Channel+Spatial module). In our experiment, the module after

conv3_x (P_2) reached 0.9510 and the one after conv5_x (P_4) 0.9310.

12. Why can spatial attention be useful at an intermediate ResNet stage?
Answer: At 28×28 or 14×14 the map is still fine enough to separate object from background, and the features are already semantic enough to recognise clutter. Down-weighting it before the deep stages is therefore possible. This is consistent with our best model being Channel+Spatial at the intermediate position P_2.

13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
Answer: A has N^2 entries ($N = HW$), so computation is $O(N^2C)$ and memory $O(N^2)$. At P_1, $N = 3136$: A has 9.8 M entries (39 MB per image, about 1.26 GB per batch of 32), and the module needs ≈ 3.1 GMAC, i.e. +75 % of ResNet50.

14. What happens to the size of the self-attention matrix when the spatial resolution increases?
Answer: It grows quadratically with N, i.e. $16\times$ for every doubling of H and W: 2,401 entries at P_4, 38,416 at P_3, 614,656 at P_2 and 9,834,496 at P_1 (Table 3).

15. How does attention performance vary when the insertion position is changed?
Answer: With the same module the accuracy was 0.9093 at P_1, 0.9510 at P_2, 0.9238 at P_3 and 0.9310 at P_4, a 4.2 pp spread. At P_1 the module re-weights edge-level features with little class meaning, and the change passes through three pretrained stages. At P_4 the 7×7 map allows only coarse decisions at a high parameter cost. P_2 balances resolution and semantics. Earlier positions also train longer (30.4 versus 19.3 min).

16. How does attention at P_2 differ visually from attention at P_4?
Answer: Across all four test images in Figure 15, $M_s^{P_2}$ (28×28) is fine-grained and nearly uniform, with a low band along the bottom border. $M_s^{P_4}$ (7×7) consists of coarse 32×32-pixel blocks that switch off whole corner regions. Resolution changes the granularity of the map, but neither map outlines the flower.

17. Does applying attention at multiple stages always improve classification performance?
Answer: No. $P_3 + P_4$ (0.9383) stayed below the baseline, and $P_2 + P_3 + P_4$ (0.8947) was the worst model of the study, even though P_2 alone was the best. Each module shifts the features that the next pretrained stage expects, and stacking compounds these shifts. It also adds 692 k parameters and 6.6 min of training, so the extra cost is not justified.

18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
Answer: Only slightly. Spatial→Channel scored 0.9328 versus 0.9238 (test loss 0.1886 versus 0.2060) with identical parameters, and the gap is within noise. Spatial-first computes its map from unscaled features, before the channel gates remove about half of the channels, which plausibly explains the small advantage.

19. How does the spatial-attention kernel size influence the attention map?
Answer: A larger kernel lets each attention value use more context (7×7 covers half of a 14×14 map), giving smoother, region-level maps; a 3×3 kernel gives more local maps. Accuracy rose monotonically (0.9056, 0.9165, 0.9238) for only 80 extra parameters, so the extra context is useful and essentially free. Only the $k = 7$ map was visualised.

20. How does SE attention differ from ECA attention?
Answer: SE uses a fully connected bottleneck $C \to C/r \to C$ with global channel interaction ($\approx 2C^2/r$, 132,160 parameters). ECA has no reduction and uses a single 1-D convolution over $k = 5$ neighbouring channels (5 parameters). ECA (0.9474) beat SE (0.9183), plausibly because the tiny module perturbs the pretrained features less and cannot overfit the small training set.

21. How does CBAM combine channel and spatial attention?
Answer: Sequentially. Channel attention comes first, built from average- *and* max-pooled descriptors passed through a shared MLP and summed before the sigmoid. Spatial attention follows, built from the channel-wise average and maximum maps with a 7×7 convolution. Our CBAM (132,259 parameters, 0.9220) performed like our Channel+Spatial module (0.9238), as expected from the near-identical design.

22. How does self-attention differ from channel attention?
Answer: Channel attention scales whole channels and does not mix positions. Self-attention mixes all positions through an $N \times N$ content-dependent matrix, giving long-range context, at a far higher cost: 1.31 M parameters and 0.30 GMAC, against 132 k and 0.33 MMAC. Ours collapsed onto one or two key positions with $\gamma = 0.066$, so it mainly added a global context vector, and its accuracy stayed near the baseline (0.9419).

23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
Answer: No. Combined (1.44 M extra parameters) was the lowest at P_3 (0.9111), self-attention (1.31 M) stayed below the baseline, and ECA (5 parameters) was the highest (0.9474). The correlation between

parameters and accuracy is −0.18. On a small dataset, extra capacity mainly widens the train–validation gap.

24. What is the difference between an internal attention map and Grad-CAM?
Answer: An internal map is computed by the module during the forward pass, changes the prediction, is class-agnostic, and exists only in attention models. Grad-CAM is computed afterwards from class-score gradients, is class-specific, leaves the model unchanged, and works for any CNN. In our results they disagree: the internal spatial maps were near-uniform (a border prior), while Grad-CAM localised the flower.

25. Why is Grad-CAM called a post-hoc explanation technique?
Answer: It is applied to an already trained model and an already made prediction, using gradients, purely to explain that prediction. It is not part of the model's computation or training.

26. Does the attention model suppress irrelevant background regions?
Answer: Not more than the baseline. Over 60 test images, every model's Grad-CAM puts little activation near the border (0.19–0.26 versus 0.44 for a uniform map). Spatial-type modules are only slightly more compact (coverage 0.18–0.20 versus 0.22), and self-attention and Coordinate attention spread more widely (Table 14). The internal spatial maps suppress only the image border, not the background.

27. Are the attention maps of correctly classified and incorrectly classified images different?
Answer: The internal maps are not: they are nearly identical for every image. The Grad-CAMs are. They lie on the flower when the prediction is correct and on background colour cues when it is wrong: pink blur under the white rose in image 222, and a pink fingertip in image 530, where the correct baseline looked at the flowers (Figures 11, 16).

28. Which flower classes benefit most from attention?
Answer: Only daisy improved (F1 +0.011, one more image). Roses lost the most (F1 −0.043), and the roses↔tulips confusion grew from 13 to 19 errors. This fits the Grad-CAM evidence that the errors follow colour cues, which attention did not learn to ignore.

29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
Answer: Which classes are confused with which, in which direction, and whether the errors are concentrated or spread out. For example, the baseline accuracy of 0.9437 hides that roses have a recall of only 0.865, with 12 of their 13 errors predicted as tulips (Figure 9).

30. Why is macro F1 useful in addition to accuracy?
Answer: It weights every class equally and balances precision and recall, so a weak class (roses) cannot hide behind strong ones. Here macro F1 tracked accuracy for every model ($|F1 - \mathrm{Acc}| \leq 0.0011$), which confirms that no mechanism traded one class for another.

31. What is meant by the accuracy–complexity trade-off?
Answer: The accuracy gained per unit of extra cost (parameters, computation, latency). A mechanism is worthwhile only if its gain exceeds the noise (≈ 3.1 pp) and the cost is acceptable. Here the efficient frontier is the baseline and ECA. Self-attention, for example, adds 1.31 M parameters and 7.4 % computation for no gain.

32. What is meant by the F1–inference-time trade-off?
Answer: The same idea for latency: does extra inference time buy a meaningful F1 improvement? Here it did not. The baseline was the fastest model (12.85 ms) and among the best, and the extra time measured for the attention models was mostly thermal throttling of the laptop rather than module cost.

33. Why must every ablation study use the same dataset split?
Answer: So that differences come from the model and not from an easier or harder test set. Sampling alone already gives ±2.2 pp of uncertainty on 551 images, and a different split per model would add variance of the same size as the effects being measured.

34. Why must the same training protocol be maintained when comparing different attention mechanisms?
Answer: Optimizer, learning rate, epochs and augmentation change accuracy as much as attention does. For example, validation accuracy moved 3–7 pp between neighbouring epochs at the start of fine-tuning. If they varied, the effect of the attention module would be confounded with them.

35. What conclusions can be drawn from the Grad-CAM comparison?
Answer: Based on 5 representative images and statistics over 60 test images: every model uses the flower when it is correct; attention did not systematically improve localisation; errors come from background colour cues; and ECA was the only model that looked at the rose in the hardest image, while also being the most accurate at P_3. These qualitative findings agree with the quantitative ones: attention gave no significant gain over the pretrained baseline.

38. Observations

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

Recorded observations

Item 1 is in Table 4. Items 2–6 are in Table 17; items 7–9 in Table 15; items 10–12 in Table 16. Items 13–18 are in Figures 9, 12, 13, 14, 11–16 and 15.

Table 17: Observations 2–6: accuracy and loss of every trained model.

Model	Pos.	Training accuracy	Validation accuracy	Testing accuracy	Training loss	Validation loss	Test loss
A0 Baseline	–	0.9988	0.9545	0.9437	0.0063	0.3077	0.1915
A1 Channel (= SE)	P_3	0.9938	0.9455	0.9183	0.0157	0.3832	0.2331
A2 Spatial	P_3	0.9977	0.9527	0.9347	0.0069	0.2802	0.1771
A3 Channel→Spatial	P_3	0.9981	0.9364	0.9238	0.0080	0.3961	0.2060
A4 Self-Attention	P_3	0.9949	0.9345	0.9419	0.0148	0.3461	0.1952
A5 Combined (C→S→SA)	P_3	0.9946	0.9455	0.9111	0.0144	0.3806	0.2710
B1 Ch→Sp	P_1	0.9988	0.9364	0.9093	0.0061	0.4588	0.2506
B2 Ch→Sp	P_2	0.9996	0.9418	0.9510	0.0040	0.2721	0.2046
B4 Ch→Sp	P_4	0.9992	0.9455	0.9310	0.0037	0.3545	0.2143
C3 Ch→Sp	P_3+P_4	0.9977	0.9455	0.9383	0.0054	0.3574	0.1922
C4 Ch→Sp	P_2+P_3+P_4	0.9977	0.9309	0.8947	0.0081	0.4597	0.3031
D1 Sp→Ch	P_3	0.9961	0.9382	0.9328	0.0094	0.4771	0.1886
E1 Ch→Sp, $k = 3$	P_3	0.9930	0.9236	0.9056	0.0214	0.5227	0.2681
E2 Ch→Sp, $k = 5$	P_3	0.9981	0.9436	0.9165	0.0072	0.4182	0.2386
F1 ECA	P_3	0.9984	0.9491	0.9474	0.0063	0.3114	0.1943
F2 CBAM	P_3	0.9973	0.9418	0.9220	0.0114	0.3332	0.2642
F3 Coordinate	P_3	0.9852	0.9345	0.9129	0.0397	0.5479	0.2870

39. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity}.\end{aligned}$$

Students should also understand that classification performance alone does not provide a complete explanation

of an attention mechanism.
A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

40. Conclusion

This experiment inserted channel, spatial, channel–spatial, self-attention, SE, ECA, CBAM and Coordinate attention into an ImageNet-pretrained ResNet50 at four positions. Seventeen configurations were trained under one identical protocol on `tf_flowers` (2,569 / 550 / 551 images). The main conclusions are:

1. **Attention type:** at P_3 no attention type beat the baseline (accuracy 0.9437, macro F1 0.9436). Self-attention (0.9419), spatial attention (0.9347) and ECA (0.9474) matched it within the ≈ 3.1 pp noise level of a 551-image test set. The Combined module was clearly worse (0.9111).
2. **Attention position:** the position mattered as much as the type. The same Channel→Spatial module varied by 4.2 pp across positions. It gave the best result of the study at the intermediate position P_2 (0.9510), where features combine enough resolution with enough semantics, and a clearly worse one at the high-resolution, low-level position P_1 (0.9093).
3. **More attention is not better:** two positions ($P_3 + P_4$, 0.9383) did not beat the baseline, and three positions ($P_2 + P_3 + P_4$, 0.8947) gave the worst model. The 5-parameter ECA matched or outperformed every heavier mechanism, including self-attention with 1.31 M parameters.
4. **Order and kernel size:** Spatial→Channel was slightly better than Channel→Spatial (0.9328 versus 0.9238, within noise). A larger spatial kernel helped consistently ($k = 3/5/7$: 0.9056 / 0.9165 / 0.9238) at negligible cost.
5. **Complexity:** only self-attention added significant computation (+7.4 % at P_3, +75 % if placed at P_1). No configuration gave a gain large enough to justify its extra parameters or inference time.
6. **Qualitative behaviour:** Grad-CAM showed that all models rely on the flower when correct and on background colour cues (pink blur, a fingertip) when wrong. The internal attention was largely input-independent: near-identical channel weights for every image, spatial maps that only suppress the image border, and self-attention collapsed onto a few positions. This explains the small quantitative effects, so the quantitative, computational and qualitative analyses agree.

Hence, attention effectiveness was decided by the interaction of attention type, attention position, feature resolution, feature semantics and model complexity, not by the mechanism alone. For a strong pretrained backbone fine-tuned on a small dataset, adding attention is not automatically beneficial. Lightweight mechanisms (ECA, spatial attention) placed at an intermediate stage are the safest choice. Repeated runs with different seeds would be needed to separate the models that lie within the noise band.

41. Source Code

`https://github.com/Kanishkarane/Deep-Learning/tree/main/Attention%20Mechanisms`

42. References

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